

Car Price Prediction Using Regularized Regression

Applying Ridge, Lasso, and ElasticNet Models

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Assignment Title:	RR Car Price Prediction
Dataset:	CarPrice_Assignment.csv
Records:	205 rows x 26 attributes
Target Variable:	price (USD)
Date:	June 2026

Project Goal

Geely Auto, a Chinese automotive manufacturer, is planning its entry into the US automobile market and wants to understand which vehicle characteristics most strongly influence pricing in that market. This report develops and compares four regression approaches — ordinary Linear Regression, Ridge (L2), Lasso (L1), and ElasticNet — to model car price as a function of design, performance, and brand attributes, and to surface the variables that matter most for pricing strategy.

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1. Library Imports and Data Loading

All packages required for the analysis — pandas, numpy, matplotlib, seaborn, scikit-learn, and statsmodels — were imported at the start of the notebook. The CarPrice_Assignment.csv file was then loaded into a pandas DataFrame, and an initial check confirmed the dataset's dimensions and column names.

Table 1: Dataset Snapshot

Property	Value
Number of rows	205
Number of columns	26
Target column	price
Missing values (total)	0
Data types	8 float64, 8 int64, 10 object (string)

Table 2: Column Reference

Column	Type	Description
car_ID	Integer	Row identifier (dropped before modelling)
symboling	Integer	Insurance risk score, ranges from -3 to +3
CarName	Text	Full make and model string
fueltype	Categorical	gas or diesel
aspiration	Categorical	std or turbo
doornumber	Categorical	two or four
carbody	Categorical	convertible, hatchback, sedan, wagon, hardtop
drivewheel	Categorical	fwd, rwd, or 4wd
enginelocation	Categorical	front or rear
wheelbase, carlength, carwidth, carheight	Float	Body dimensions in inches
curbweight	Integer	Vehicle weight (lbs), excluding occupants
enginetype	Categorical	dohc, ohc, ohcv, l, rotor, etc.
cylindernumber	Categorical	two through twelve cylinders
enginesize	Integer	Engine displacement
fuelsystem	Categorical	mpfi, 2bbl, idi, etc.
boreratio, stroke	Float	Cylinder bore and piston stroke
compressionratio	Float	Engine compression ratio
horsepower, peakrpm	Integer	Engine output and peak RPM
citympg, highwaympg	Integer	Fuel economy figures
price	Float	Target: vehicle price in USD

2. Exploratory Data Analysis (EDA)

Before any modelling took place, the dataset was explored to understand its structure, the shape of the target variable, the influence of categorical attributes, and the relationships among numerical predictors.

2.1 Initial Data Inspection

A call to `df.info()` and `df.describe()` confirmed that the dataset has 205 rows across 26 columns with **zero missing values** in any field, so no imputation strategy was required. The table below summarises the central tendency and spread of the most important numerical fields.

Table 3: Summary Statistics for Selected Numerical Fields

Statistic	price	enginesize	horsepower	curbweight	citympg	highwaympg
Count	205	205	205	205	205	205
Mean	13,276.71	126.91	104.12	2555.57	25.22	30.75
Std Dev	7,988.85	41.64	39.54	520.68	6.54	6.89
Min	5,118.00	61	48	1488	13	16
25%	7,788.00	97	70	2145	19	25
Median	10,295.00	120	95	2414	24	30
75%	16,503.00	141	116	2935	30	34
Max	45,400.00	326	288	4066	49	54

2.2 Examining the Price Variable

The target variable, price, spans from \$5,118 to \$45,400 with a mean of \$13,276.71 and a median of \$10,295.00. Because the mean sits well above the median, the distribution leans toward lower prices with a long tail of premium vehicles; the computed skewness of approximately **1.78** confirms this positive skew. A `log1p` transformation was applied alongside the raw distribution to illustrate how the skew is reduced under a logarithmic scale.

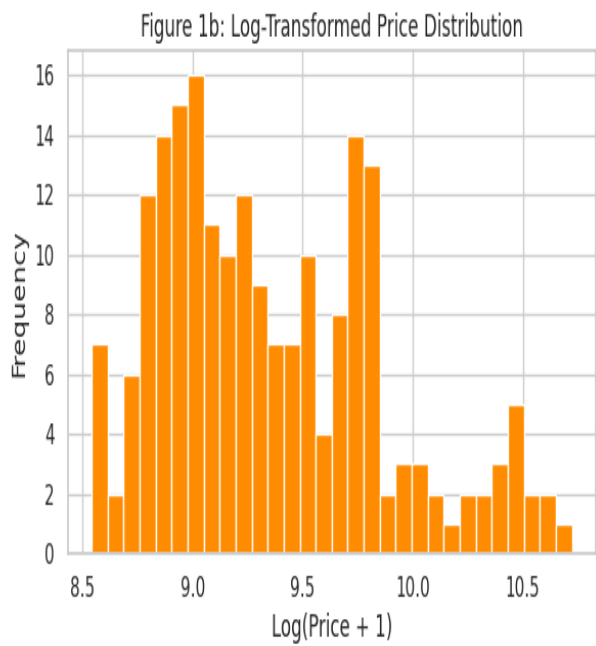
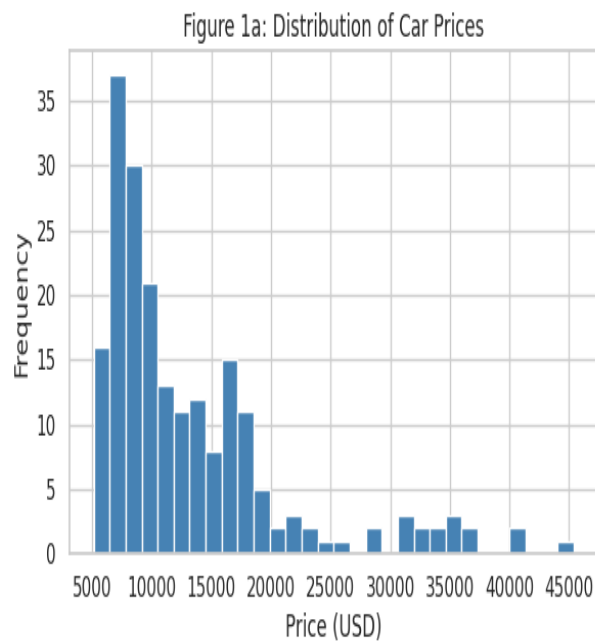


Figure 1: Raw and log-transformed distributions of car price

2.3 Deriving the Car Manufacturer Column

The CarName field combines the manufacturer and model into a single string (e.g. "alfa-romero giulia"). A new carcompany column was derived by taking the first token of CarName and lower-casing it. Several entries contained spelling inconsistencies — maxda, toyouta, vokswagen, vw, porcshce, and alfa-romero — which were corrected to mazda, toyota, volkswagen, volkswagen, porsche, and alfa-romeo respectively via a lookup dictionary. The cleaned column yields **22 distinct manufacturers**, with Toyota (32), Nissan (18), and Mazda (17) contributing the most records.

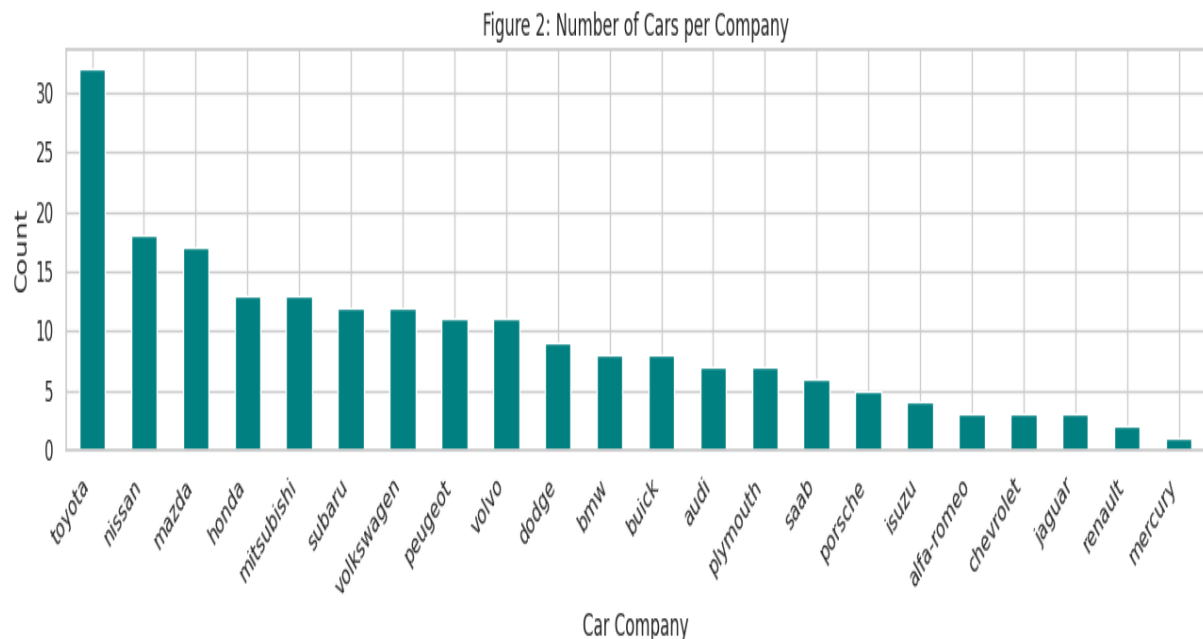


Figure 2: Count of vehicles listed per manufacturer

2.4 Categorical Attributes vs Price

For each of six categorical columns — fueltype, aspiration, doornumber, carbody, drivewheel, and enginelocation — the mean price was computed per category and plotted. The charts surface several patterns useful for downstream interpretation:

- **Fuel type:** diesel-powered vehicles average a higher price than gas-powered ones.
- **Aspiration:** turbocharged engines are associated with a notable price premium over standard aspiration.
- **Door number:** two-door configurations skew toward higher average prices, reflecting sportier body styles.
- **Car body:** hardtops and convertibles occupy the top of the price range, while hatchbacks sit at the lower end.
- **Drive wheel:** rear-wheel-drive vehicles command substantially higher average prices than front-wheel-drive ones.
- **Engine location:** rear-engine placement is associated with the highest average price band, consistent with exotic/sports configurations.

Figure 3: Average Car Price by Categorical Features

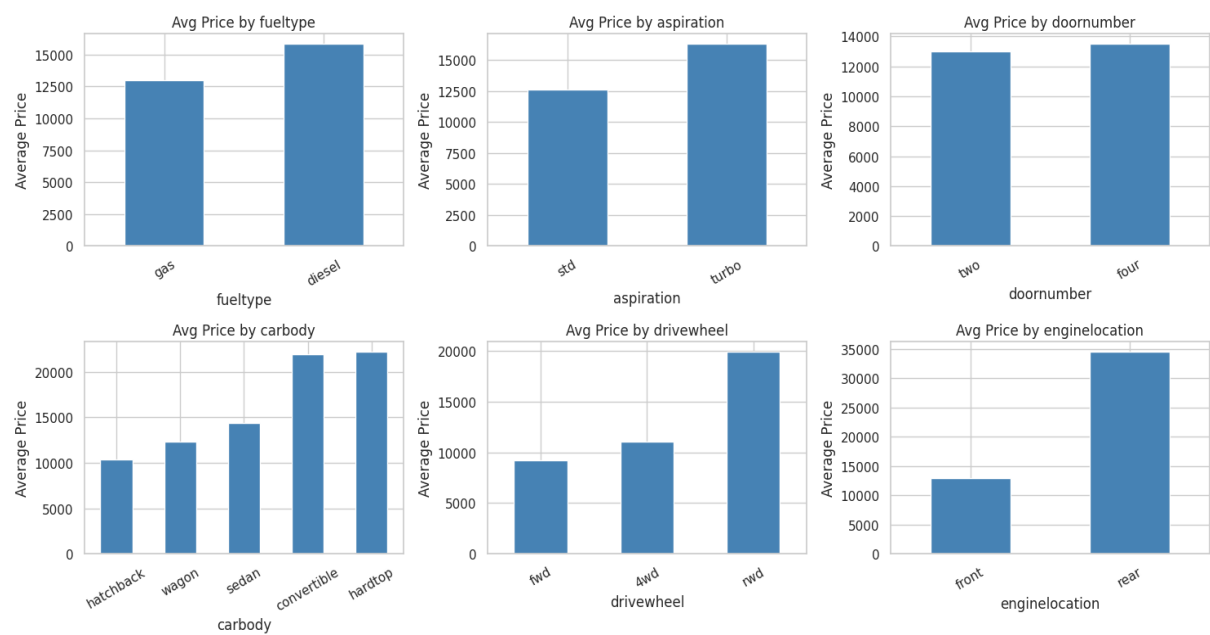


Figure 3: Mean price across categories of six categorical attributes

2.5 Numerical Feature Correlations

A Pearson correlation matrix was computed across 15 numerical columns, including price, and visualised as a lower-triangular heatmap. The six attributes with the largest absolute correlation with price were then identified for closer inspection via scatter plots.

Table 4: Correlation of Numerical Attributes with Price

Feature	Correlation (r)	Direction
enginesize	0.874	Strong positive
curbweight	0.835	Strong positive
horsepower	0.808	Strong positive
carwidth	0.759	Strong positive
highwaympg	-0.698	Strong negative
citympg	-0.686	Strong negative
carlength	0.683	Moderate positive
wheelbase	0.578	Moderate positive
boreratio	0.553	Moderate positive
carheight	0.119	Negligible
symboling	-0.080	Negligible
stroke	0.079	Negligible
compressionratio	0.068	Negligible

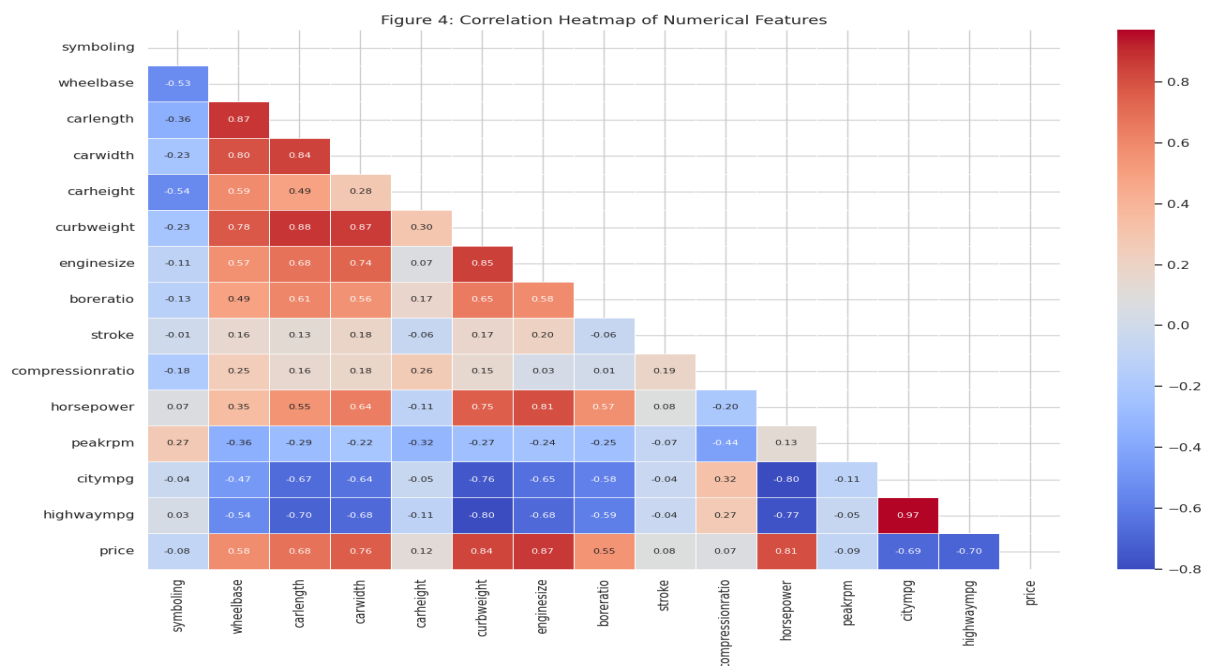


Figure 4: Correlation heatmap of numerical attributes (lower triangle)

The scatter plots below confirm the directionality found in the correlation table: engine size, curb weight, horsepower, and car width all rise together with price, while the two fuel-economy measures (city and highway mpg) fall as price increases — larger, more powerful vehicles tend to consume more fuel and cost more.

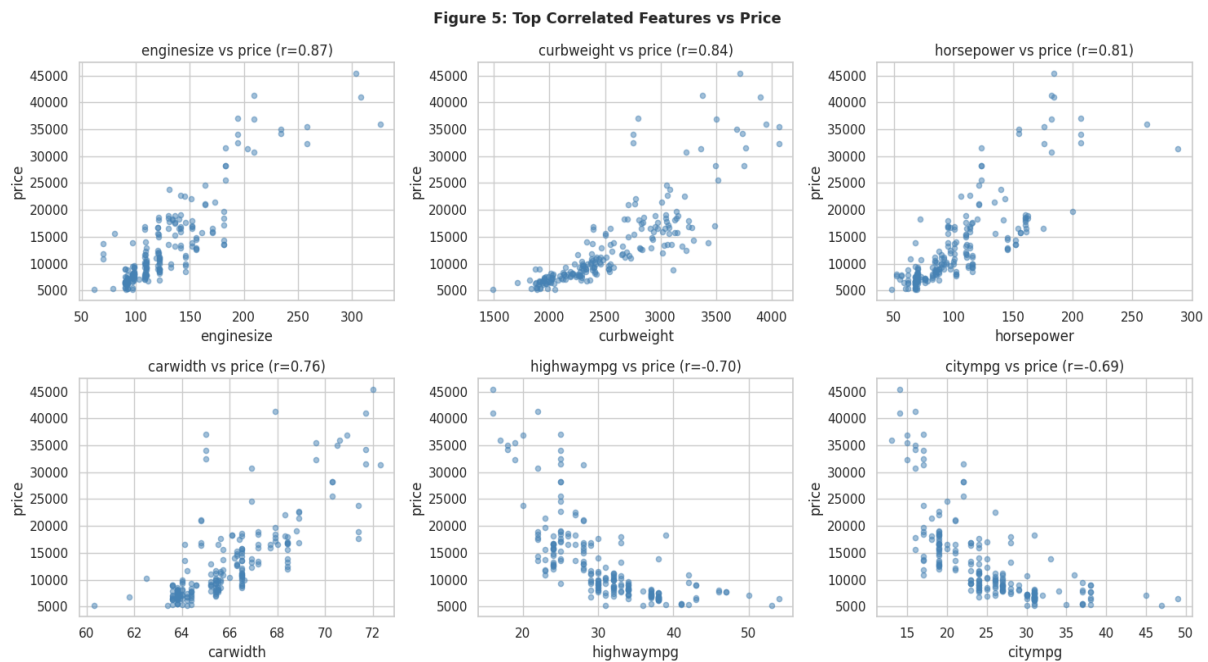


Figure 5: Scatter plots of the six most price-correlated features

2.6 Identifying Outliers in Key Features

Box plots were drawn for the same six top-correlated features to flag potential outliers. Engine size and horsepower in particular show a handful of high-value points that sit well above the upper whisker; these correspond to high-performance and luxury vehicles rather than data-entry errors, and were therefore **kept in the dataset** for modelling.

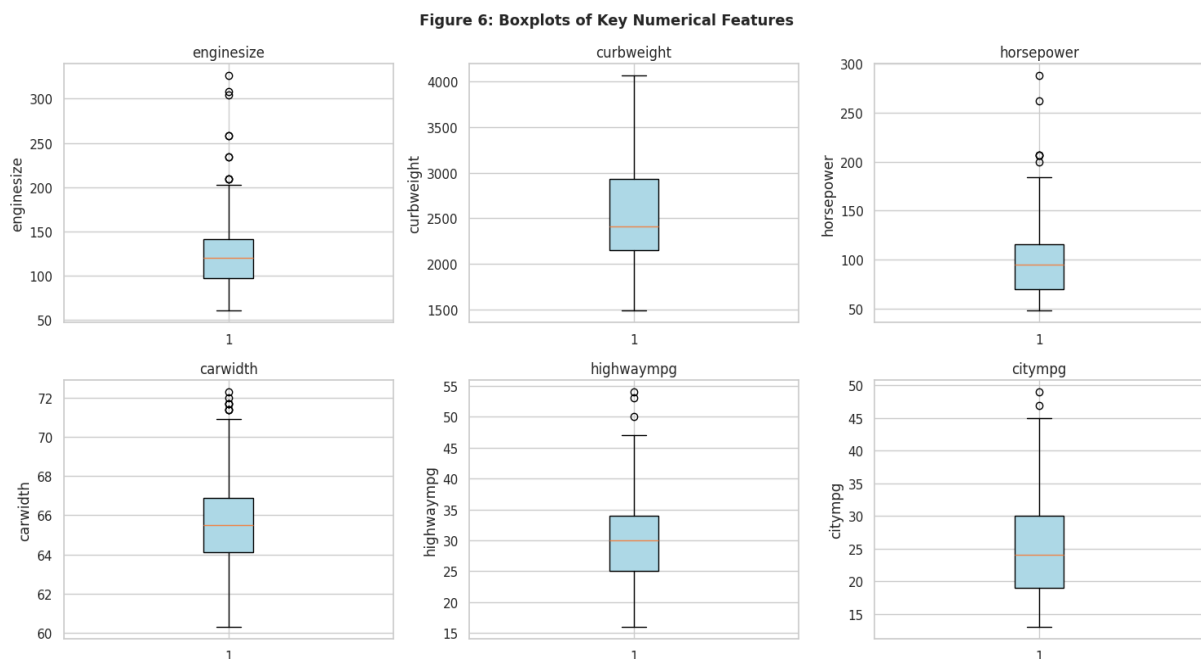


Figure 6: Box plots of the six most price-correlated numerical features

3. Data Preprocessing

Before fitting any regression model, the raw, partly categorical dataset was converted into a fully numeric feature matrix, split into training and test partitions, and scaled.

3.1 Encoding Categorical Variables

Two encoding strategies were used depending on how many distinct values a categorical column held:

Two-level columns were mapped directly to 0/1 using a dictionary (fueltype: gas=1/diesel=0, aspiration: turbo=1/std=0, doornumber: two=1/four=0, enginelocation: front=1/rear=0).

Higher-cardinality columns — carbody, drivewheel, enginetype, cylindernumber, fuelsystem, and carcompany — were one-hot encoded with the first category dropped (drop_first=True) to avoid the dummy-variable trap.

Table 5: One-Hot Encoding Expansion

Column	Distinct Values	Dummy Columns Created
carbody	5	4
drivewheel	3	2
enginetype	7	6
cylindernumber	7	6
fuelsystem	8	7
carcompany	22	21

After dropping car_ID and CarName and applying both encodings, the modelling table grew from 26 to **65 columns** (64 features plus the price target), with no missing values introduced by the transformation.

3.2 Data Splitting and Standardization

The 65-column table was split into a feature matrix X (64 columns) and target vector y (price). An 80/20 train-test split with random_state=42 produced **164 training rows** and **41 test rows**. A StandardScaler instance was fitted exclusively on the training rows (so it learns its mean and variance only from training data) and then used to transform both the training and test feature matrices, ensuring no information from the test set leaks into the scaling step.

Table 6: Split and Scaling Configuration

Parameter	Value
Total rows	205
Training rows	164 (80%)
Test rows	41 (20%)
Random seed	42
Feature columns	64
Scaler	StandardScaler (fit on train, applied to train & test)

4. Multicollinearity Diagnosis - Variance Inflation Factor (VIF)

Highly correlated predictors can destabilise ordinary least-squares coefficient estimates, inflating their variance and making them difficult to interpret. The Variance Inflation Factor (VIF) was computed for the 14 base numerical attributes (prior to one-hot encoding) to flag this issue. A VIF above 10 is generally treated as evidence of problematic multicollinearity.

Table 7: VIF Scores for Numerical Attributes

Feature	VIF	Interpretation
citympg	25.86	Severe - exceeds threshold
highwaympg	20.18	Severe - exceeds threshold
curbweight	15.46	High - exceeds threshold
carlength	8.79	Moderate
wheelbase	8.42	Moderate
horsepower	7.83	Acceptable
enginesize	6.70	Acceptable
carwidth	5.05	Acceptable
compressionratio	2.27	Low
carheight	2.25	Low
peakrpm	2.11	Low
boreratio	1.97	Low
symboling	1.95	Low
stroke	1.18	Low

Three attributes — **citympg**, **highwaympg**, and **curbweight** — exceed the VIF=10 threshold, reflecting the well-known relationship between vehicle weight, engine demands, and fuel economy. This redundancy among predictors is one of the main motivations for evaluating Ridge, Lasso, and ElasticNet alongside plain OLS, since the penalty terms in these models constrain coefficient magnitudes and reduce sensitivity to correlated inputs.

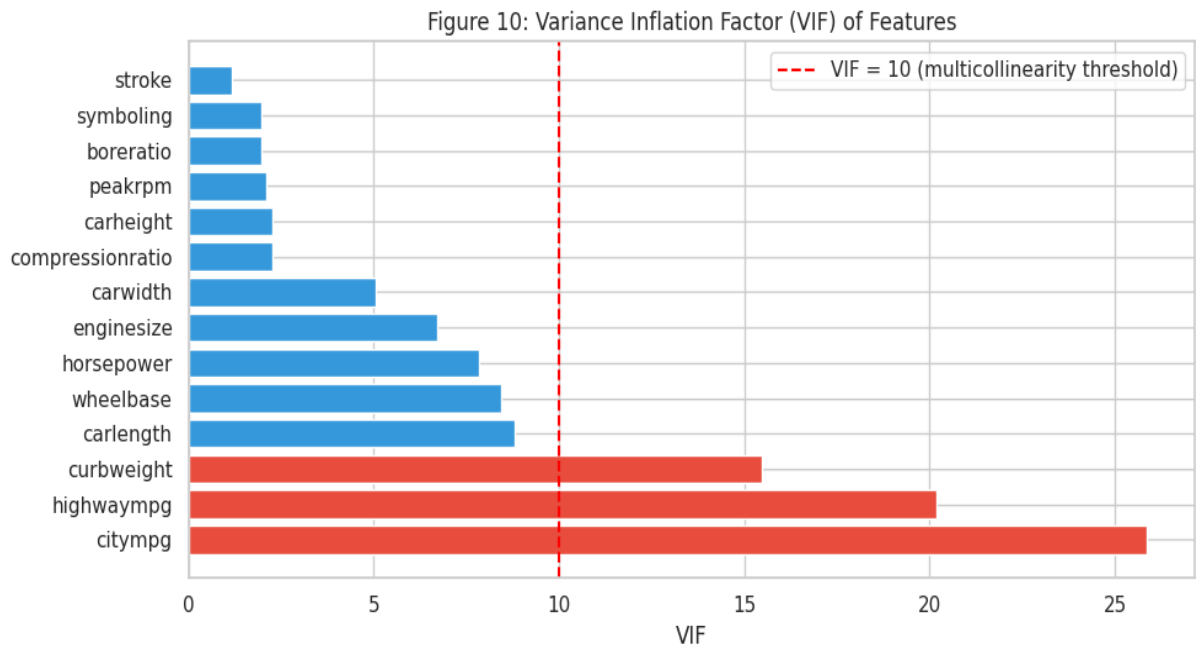


Figure 10: VIF scores for numerical attributes (red bars exceed the VIF=10 threshold)

5. Regression Model Training

Four regression models were trained on the standardized 64-feature training matrix: an ordinary Linear Regression baseline, and three regularized variants (Ridge, Lasso, and ElasticNet). Each was evaluated on the same 41-row held-out test set.

5.1 Ordinary Linear Regression - Baseline

A plain OLS model was fitted first to establish a reference point. It minimises the sum of squared residuals with no constraint on coefficient size:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2$$

On the test set, this baseline achieved an RMSE of **\$2,669.93**, an MAE of **\$1,763.57**, and an R^2 of **0.9097**.

5.2 Ridge Regression (L2 Penalty)

Ridge regression adds a penalty proportional to the sum of squared coefficients, which shrinks all coefficients toward (but not exactly to) zero:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \alpha * \sum (\beta_j^2)$$

The regularization strength α was chosen by 5-fold cross-validation over the grid [0.01, 0.1, 1, 10, 50, 100, 200, 500, 1000]. The best mean cross-validated R^2 (**0.9153**) was obtained at **$\alpha = 10$** . Refitting Ridge with this α on the full training set and evaluating on the test set gave an RMSE of **\$2,928.00**, an MAE of **\$1,984.33**, and an R^2 of **0.8914**.

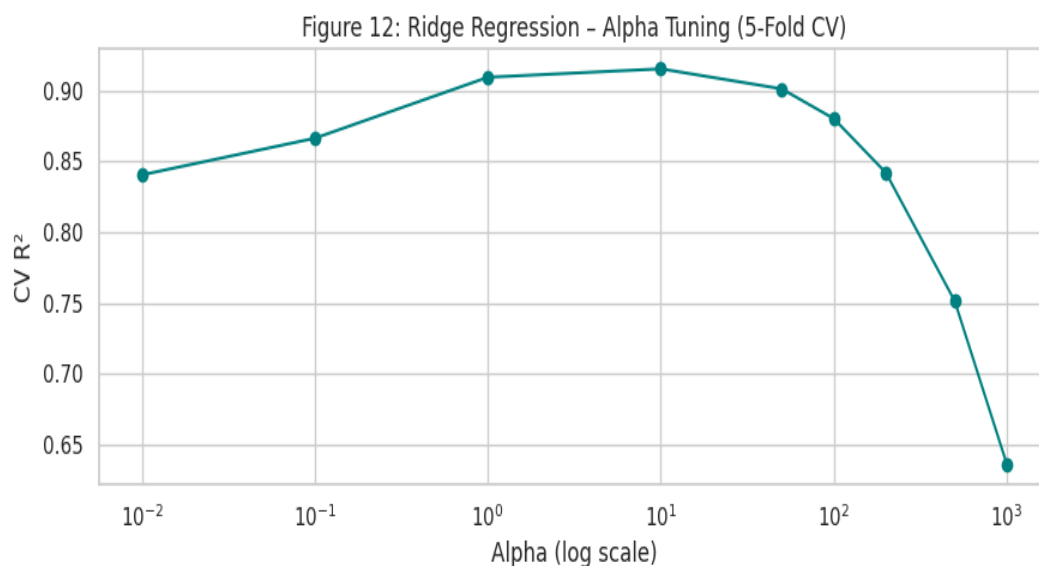


Figure 12: Mean 5-fold cross-validated R -squared across candidate Ridge α values

5.3 Lasso Regression (L1 Penalty)

Lasso regression replaces the squared penalty with an absolute-value penalty, which can drive individual coefficients exactly to zero:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \alpha * \sum |\beta_j|$$

With $\alpha = 50$, the fitted Lasso model produced an RMSE of **\$3,001.34**, an MAE of **\$2,090.68**, and an R^2 of **0.8859** on the test set. Of the 64 candidate coefficients, **25 were driven exactly to zero** and **39 remained**

non-zero, demonstrating Lasso's built-in feature-selection behaviour. The chart below shows the 20 largest non-zero coefficients by magnitude.

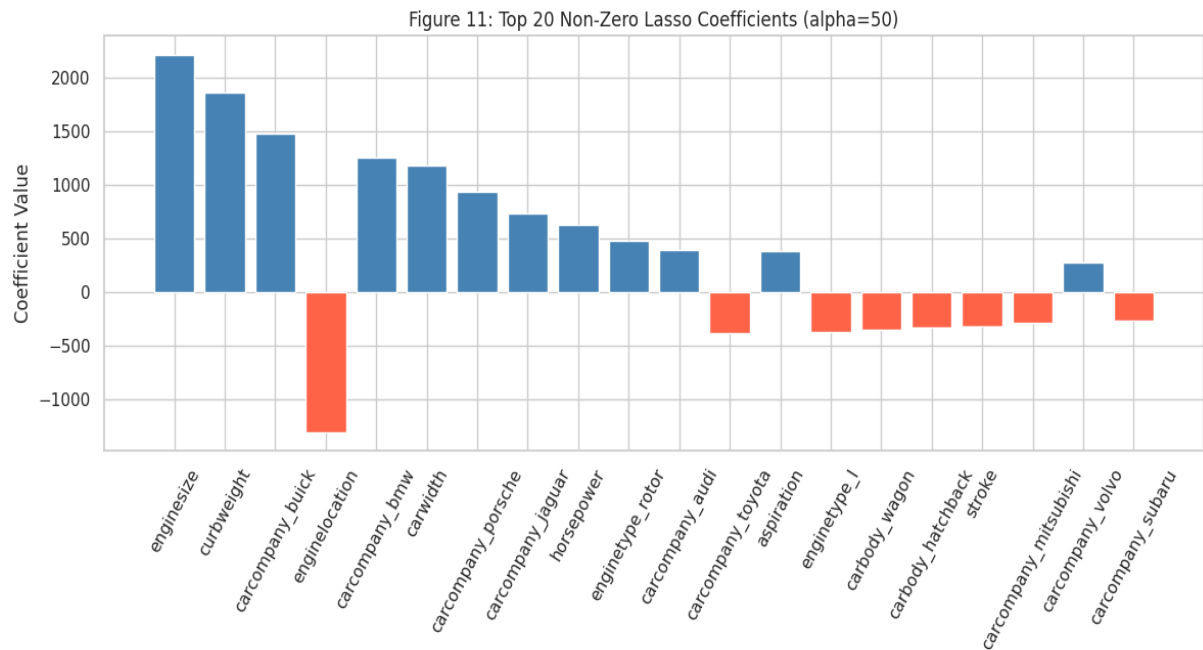


Figure 11: Top 20 non-zero Lasso coefficients (alpha=50); blue bars raise price, red bars lower it

5.4 ElasticNet - Combined L1 + L2 Penalty

ElasticNet blends the Ridge and Lasso penalties through a mixing parameter `l1_ratio`:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \alpha * [l1_ratio * \sum |\beta_j| + (1 - l1_ratio) * \sum (\beta_j^2)]$$

With $\alpha = 50$ and `l1_ratio = 0.5` (an equal blend of the two penalties), the model produced an RMSE of **\$7,000.77**, an MAE of **\$4,862.51**, and an R^2 of only **0.3792** on the test set — substantially worse than the other three models. This suggests the combined penalty at $\alpha = 50$ is too aggressive for this feature set, and that a dedicated grid search over both α and `l1_ratio` would likely be required to bring ElasticNet in line with Ridge and Lasso.

6. Model Comparison and Evaluation

All four fitted models were applied to the same 41-row test set, and three standard regression metrics — RMSE, MAE, and R^2 — were computed for each.

Table 8: Final Test-Set Performance

Model	RMSE (USD)	MAE (USD)	R-squared	Rank
Linear Regression	\$2,669.93	\$1,763.57	0.9097	1st
Ridge (alpha=10)	\$2,928.00	\$1,984.33	0.8914	2nd
Lasso (alpha=50)	\$3,001.34	\$2,090.68	0.8859	3rd
ElasticNet (alpha=50)	\$7,000.77	\$4,862.51	0.3792	4th

Table 9: Metric Definitions

Metric	Formula	What it measures
RMSE	$\sqrt{\sum (y_i - \hat{y}_i)^2 / n}$	Average error in USD, with larger errors penalised more heavily
MAE	$\sum y_i - \hat{y}_i / n$	Average absolute error in USD; less sensitive to a few large misses
R-squared	$1 - SS_{\text{res}} / SS_{\text{tot}}$	Share of price variance explained by the model (1.0 = perfect fit)

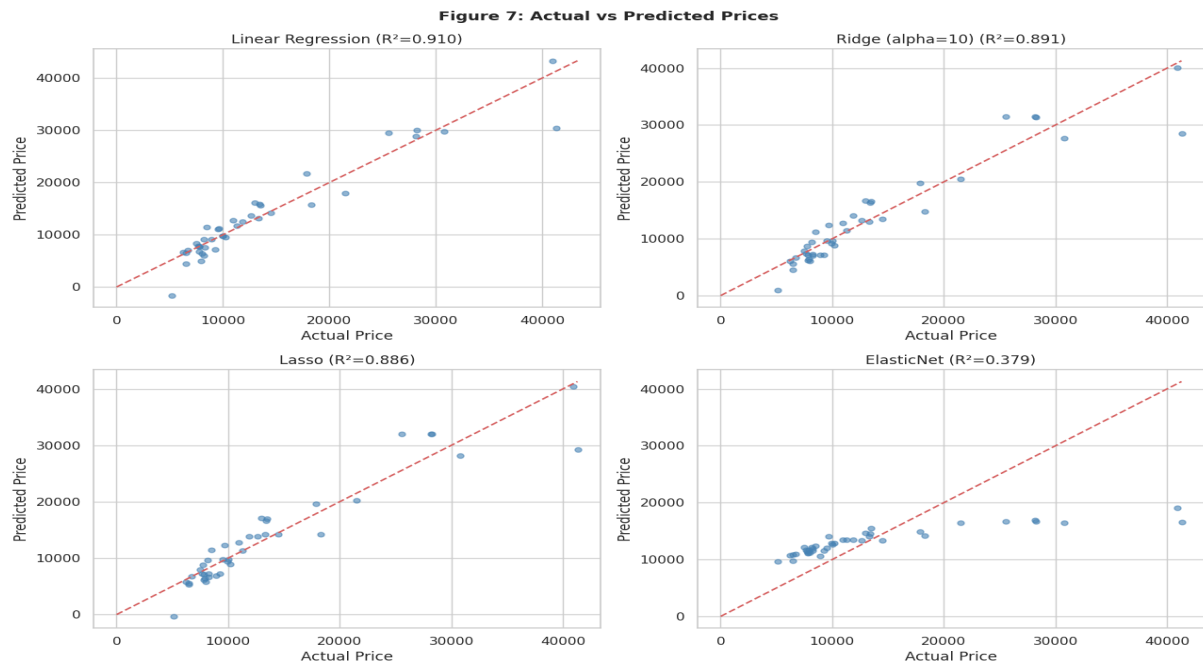


Figure 7: Predicted vs actual price for each model on the test set (dashed line = perfect prediction)

Figure 8: Residual Plots

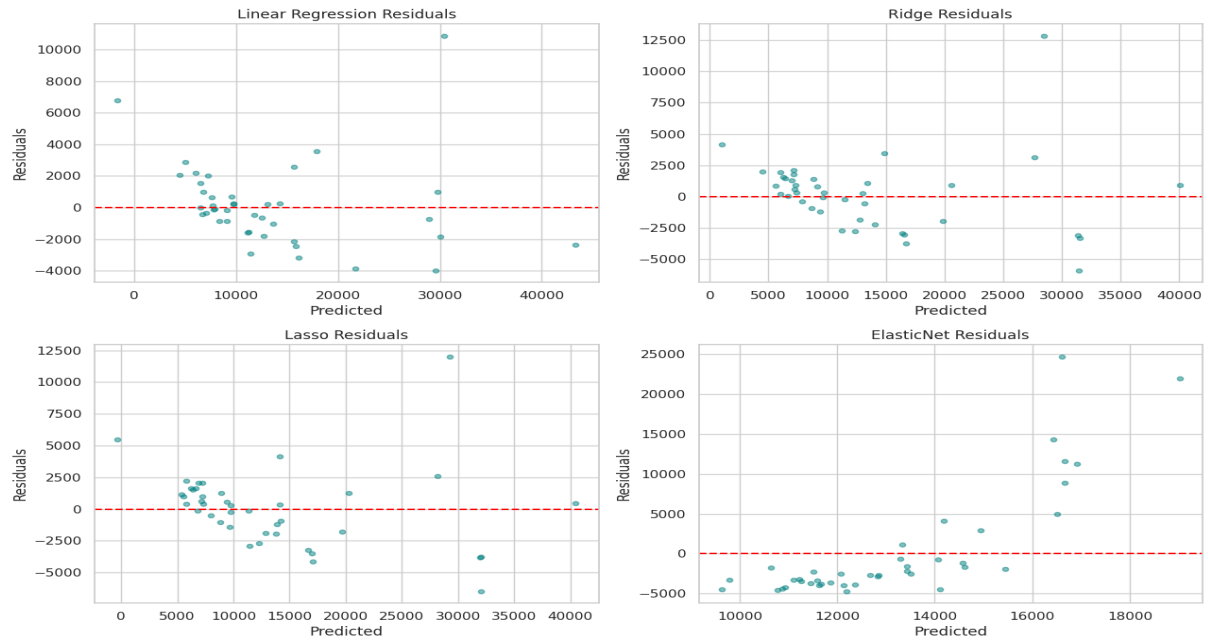


Figure 8: Residuals (actual - predicted) plotted against predicted price for each model

Figure 9: Model Performance Comparison

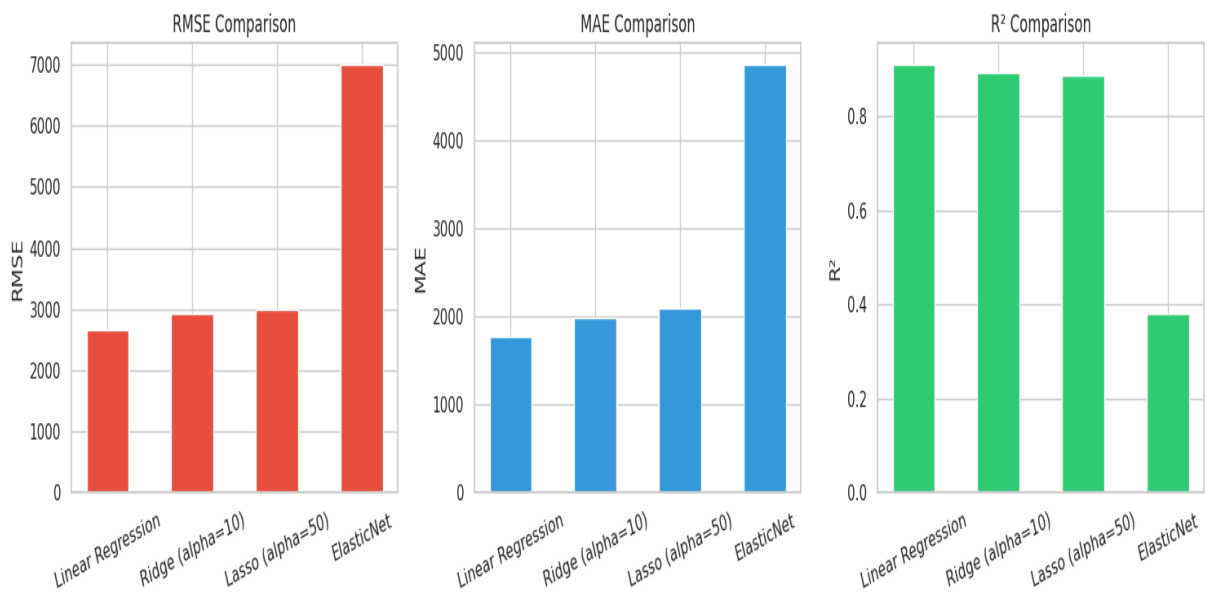


Figure 9: Side-by-side comparison of RMSE, MAE, and R-squared across all four models

7. Results, Insights, and Conclusions

7.1 Key Findings

- **Target distribution:** Car prices range from \$5,118 to \$45,400 with a mean of \$13,276.71, a median of \$10,295.00, and a skewness of approximately 1.78 — the bulk of vehicles are modestly priced, with a smaller number of premium models pulling the average upward.
- **Strongest positive predictors:** enginesize ($r=0.874$), curbweight ($r=0.835$), horsepower ($r=0.808$), and carwidth ($r=0.759$) are the numerical attributes most strongly associated with higher prices — larger, more powerful cars cost more.
- **Strongest negative predictors:** highwaympg ($r=-0.698$) and citympg ($r=-0.686$) move in the opposite direction to price — more fuel-efficient cars tend to be cheaper.
- **Brand effect:** Manufacturer matters. Premium marques such as Jaguar, Buick, and Porsche carry average prices well above the dataset mean, even after accounting for specification differences.
- **Multicollinearity:** citympg, highwaympg, and curbweight each exceed a VIF of 10, confirming that several numerical attributes are redundant with one another and motivating the use of regularized models.
- **Best performing model:** Ordinary Linear Regression achieved the best test-set performance overall ($R^2=0.9097$, $RMSE=\$2,669.93$), narrowly ahead of Ridge ($R^2=0.8914$) and Lasso ($R^2=0.8859$).
- **Feature selection via Lasso:** At $\alpha=50$, Lasso zeroed out 25 of 64 coefficients while retaining $R^2=0.8859$, offering a more interpretable model with only a small drop in accuracy compared to the full OLS fit.
- **ElasticNet underperformance:** At $\alpha=50$ and $l1_ratio=0.5$, ElasticNet trailed all other models ($R^2=0.3792$), indicating this configuration over-penalises the model; further tuning of α and $l1_ratio$ is recommended before drawing conclusions about ElasticNet's suitability.

7.2 Modelling Assumptions

- The dataset contained no missing values, so no imputation was performed.
- The car brand was taken to be the first whitespace-delimited token of CarName, lower-cased and corrected for known spelling variants.
- Two-category attributes were label-encoded to 0/1; higher-cardinality categorical attributes were one-hot encoded with the first dummy dropped to avoid the dummy-variable trap.
- StandardScaler was fitted only on the training partition and then applied to both the training and test partitions, to avoid information leakage from the test set.
- An 80/20 train-test split with $random_state=42$ was used consistently for every model, so that all four models are compared on an identical test set.
- Outliers identified in enginesize and horsepower were treated as genuine high-performance vehicles rather than data errors, and were retained in the modelling data.
- Ridge's α was tuned via 5-fold cross-validation on the training set only; Lasso and ElasticNet were run at a fixed α of 50, and ElasticNet at a fixed $l1_ratio$ of 0.5, as a starting point rather than a fully tuned configuration.
- All monetary figures are reported in US dollars, as provided in the source data, with no currency conversion applied.

7.3 Recommendations for Geely Auto

- Prioritise enginesize, curbweight, horsepower, and carwidth as the primary engineering levers when positioning a new model within a target price band.
- Consider a log-transformation of price as the regression target in future iterations, given its positive skew, and re-evaluate whether this improves residual behaviour.
- Investigate the citympg / highwaympg / curbweight cluster more closely; combining or removing one of these correlated measures could simplify the model without much loss of accuracy.
- Re-run ElasticNet with a joint grid search over alpha and l1_ratio before drawing conclusions about its competitiveness against Ridge and Lasso.
- Use the Lasso-selected feature subset (39 non-zero coefficients) as a starting point for a simplified, more interpretable pricing model for stakeholder communication.

8. References

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 - [9] CarPrice_Assignment.csv dataset, sourced via GitHub repository (SampattKumar/Linear-Regression-Car-Dataset)

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